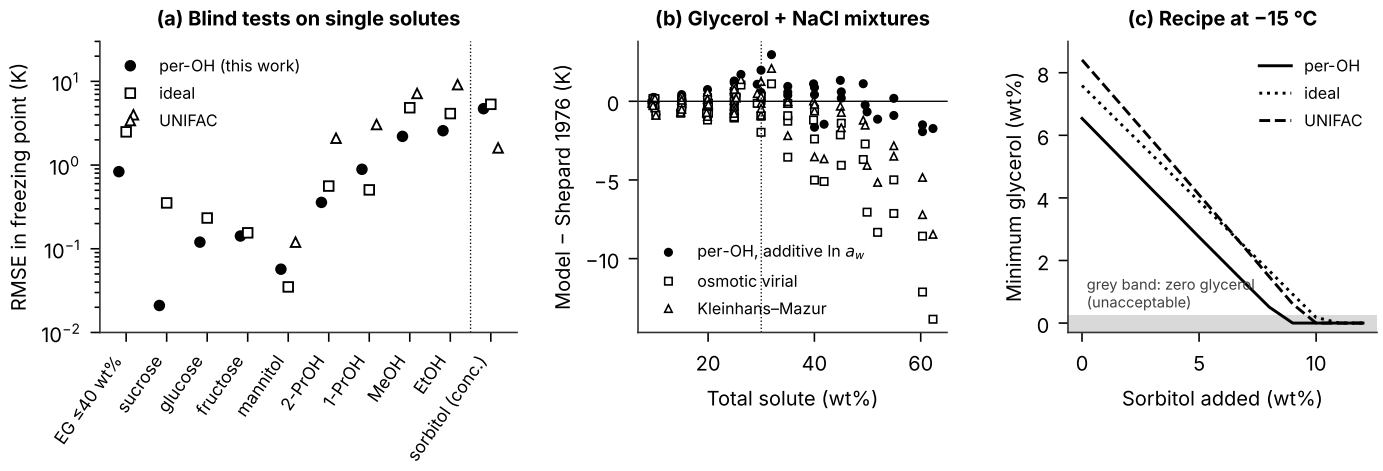
(a) Blind tests on single solutes
RMSE in freezing point (K)
per-OH (this work)
ideal
UNIFAC
EG ≤40 wt%
sucrose
glucose
fructose
mannitol
2-PrOH
1-PrOH
MeOH
EtOH
sorbitol (conc.)
(b) Glycerol + NaCl mixtures
Model − Shepard 1976 (K)
per-OH, additive ln a_w
osmotic virial
Kleinhans–Mazur
Total solute (wt%)
(c) Recipe at −15 °C
Minimum glycerol (wt%)
per-OH
ideal
UNIFAC
grey band: zero glycerol (unacceptable)
Sorbitol added (wt%)